Explainable artificial intelligence

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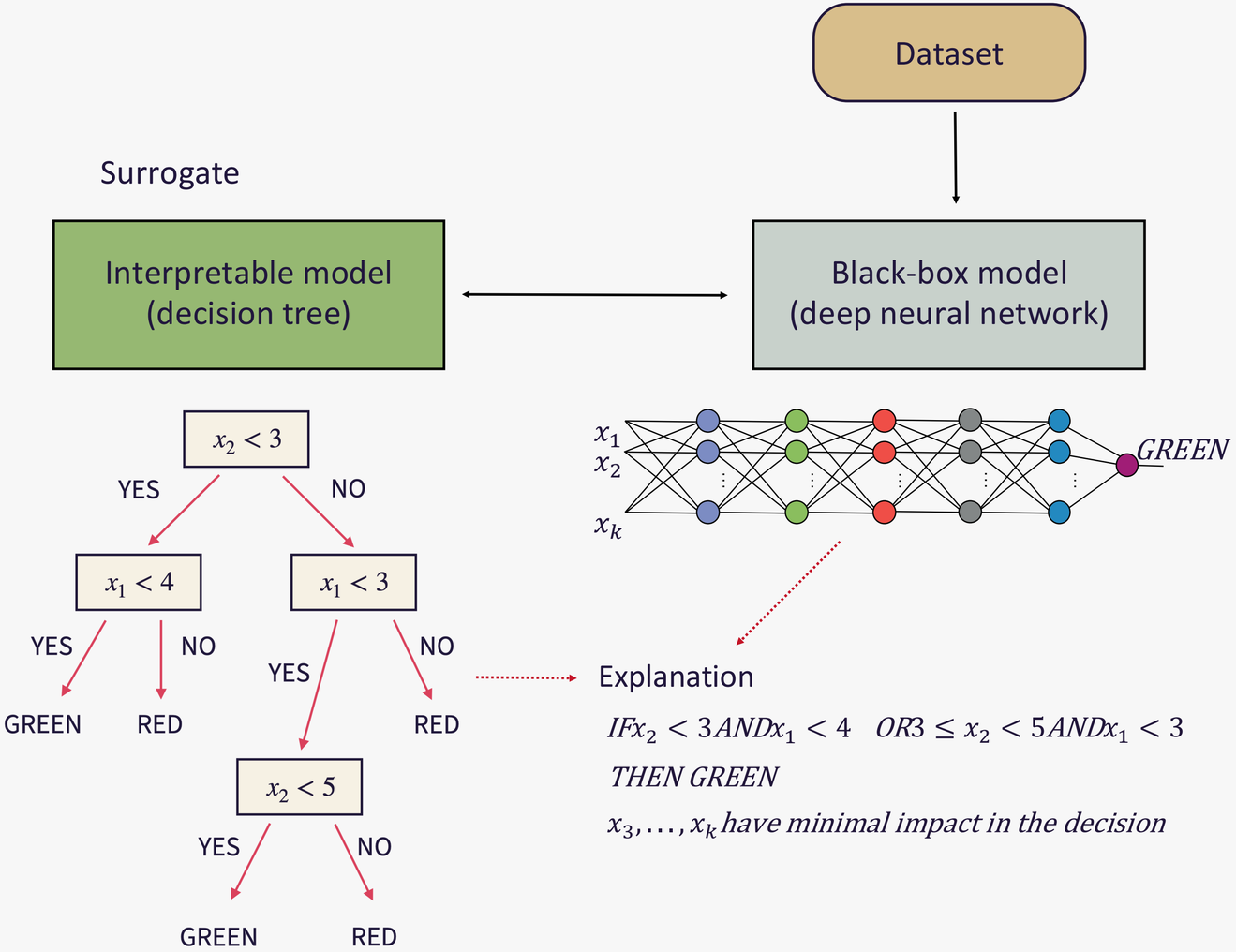
Global explanation of AI predictors - Interpretable surrogate models

Global explanation

- Provide insights into the general behavior of the model, its decision rules, the influence of different features, and potential biases
  - "I classify instances as Category X when Feature A is high and Feature B is low. However, when Feature A is moderate but Feature C is high, I classify instances as Category Y."
  - "Overall, my predictions are most influenced by Feature A, Feature B, and Feature C. Feature A contributes to 40% of the decisions, Feature B to 30%, and Feature C to 20%. Other features have minimal impact."

Methodology

- Mimic the black-box model: approximate the behavior of a black-box model (a deep neural network) by using an interpretable surrogate model (a decision tree)



Local explanation of AI predictors - Local interpretation using surrogate models

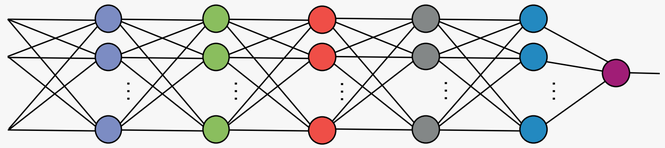
Local interpretation

- Explain why a specific instance was given a particular prediction by an AI model; understand what features contributed to the decision
  - "The loan was rejected mainly because of the applicant's low credit score (650) and high debt-to-income ratio (45%). The part-time employment status also negatively impacted the decision."

Methodology

- Local Interpretable Model-agnostic Explanations (LIME): Perturb the input; observe prediction changes; train a surrogate model that approximates the black-box model for this instance; extract feature importance for this decision

image created by DALL·E



Surrogate

Interpretable model (decision tree)

small rounded ears

long and striped tail

bold black stripes on an orange coat, muscular body

large, golden-yellow or amber-colored eyes with round pupils

"Instance A was classified as a tiger because it has bold black stripes on an orange coat, a muscular body, a long and striped tail, small rounded ears, large, golden-yellow or amber-colored eyes with round pupils"

Reading
Ribeiro, Marco Tulio et al." Why Should I Trust You? " Explaining the Predictions of Any Classifier.", KDD 2016

Counterfactual explanation

Inspecting counterfactuals

- Inspect the minimal changes to an input would lead to a different prediction by an AI model; Instead of explaining why an AI model made a decision, show what changes would alter the decision
  - "The loan was rejected but if your credit score increased from 650 to 700 and your income rose from $50,000 to $60,000, your loan would be approved."
  - "If the same cat were in a living room, the model would classify it as a domestic cat instead of a tiger."

Methodology

- Optimization: Identify the smallest feature modifications required to reverse the prediction
- Prototypes: Find a real-world example that belongs to a different class; generative models are employed to generate realistic counterfactual instances

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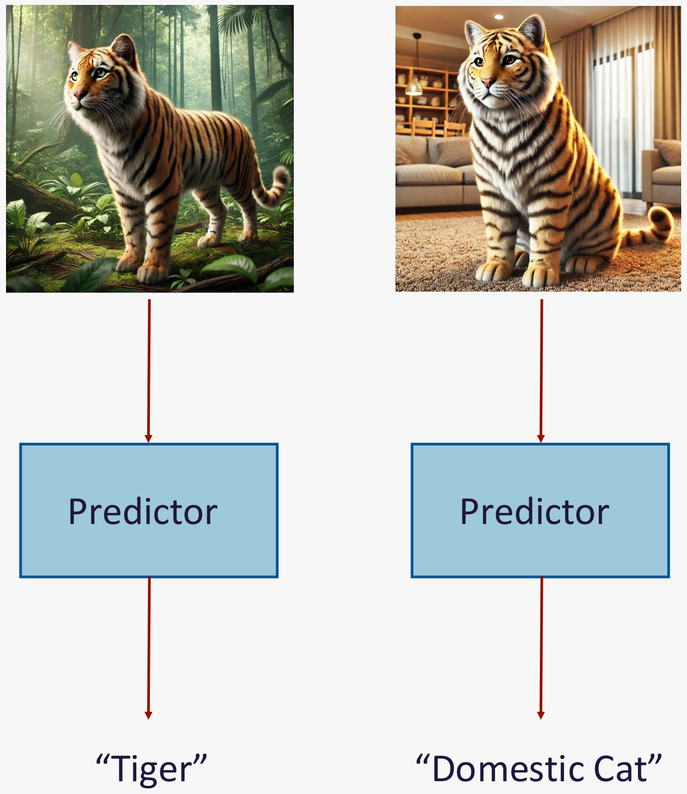

Changing the background alters the classification outcome

Take home message

- AI is useful, but it comes with risks
  - Artificial Intelligence is useful in practice across numerous domains, but it can cause both material and immaterial harm
- Due to the risks, humans don't trust opaque AI systems
  - Algorithmic aversion can be addressed by increasing AI transparency, not only accuracy
- Explanations can help making AI systems transparent
  - Extracting explanations can clarify every step of the development and use of AI, reducing the influence of the black-box effect in the lack of trust
- Providing intuitive and useful explanations is not straight forward
  - The context in which explanations are provided is often unclear
  - Different forms of explanations are useful for different applications
- Machine learning interpretation is a challenging problem
  - From a technical point of view, extracting meaningful and accurate explanations is challenging; several methods have been proposed and used in practice — there is room for improvement